



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

- The program is designed to be a simple error/incident ticket management system. Basically it manages incident tickets using a list, allowing the user to add, display, search, remove, and count tickets through a menu.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
if choice == "1":  
    tickets.append([input("Incident ID: "), input("Bot: "),  
input("Description: ")])
```

If the user chooses 1, the program asks for three things, “Incident ID”, “Bot”, and “Description”. Then, tickets.append adds them to the list.

Display Incidents:

```
elif choice == "2":  
    for t in tickets:  
        print(t[0], "|", t[1], "|", t[2])
```

The for loop goes through every ticket. The ‘t’ in “for t in tickets:” represent the current ticket. Then, ‘t[0]’ gets the Incident ID, ‘t[1]’ gets the Bot name, and ‘t[2]’ gets the Description.



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Search Incident:

```
elif choice == "3":  
    id = input("Incident ID: ")  
    for t in tickets:  
        if t[0] == id:  
            print("Found:", t[0], "|", t[1], "|", t[2])  
            break
```

The user enters an Incident ID, then the program checks every ticket. It compares the ticket's ID with the ID entered by the user

Remove Resolved Incident:

```
elif choice == "4":  
    id = input("Incident ID to remove: ")  
    for i in range(len(tickets)):  
        if tickets[i][0] == id:  
            tickets.pop(i)  
            print("Removed.")  
            break
```

First, the user enters the ID they want to remove, then “for i in range(len(tickets)):” loops through the indexes of the list. If it matches, “ticket.pop(i)” removes that ticket. If no matching ticket is found, “print(“not found”)” is executed.

Count Active Incidents:

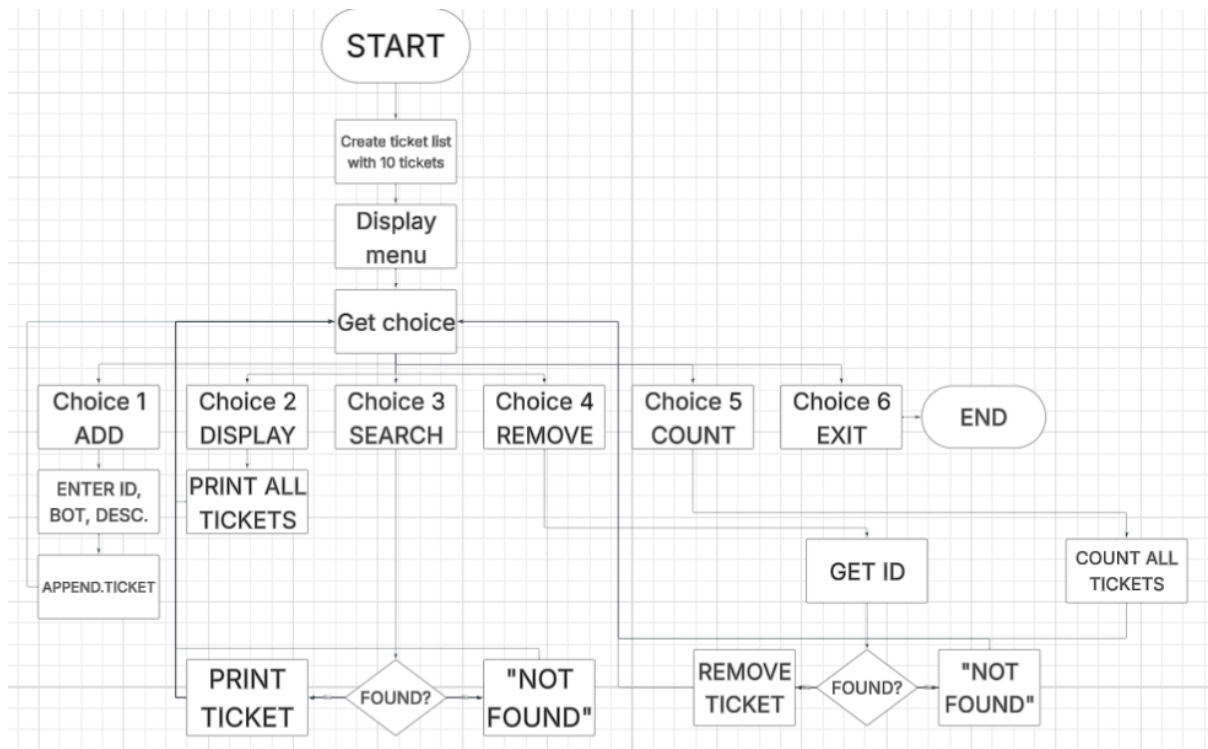
```
elif choice == "5":  
    print("Active tickets:", len(tickets))
```

len(tickets) tells you how many tickets are currently in the list.



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III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

I learned to use Python lists, loops, conditions, and list operations to design a ticket management system. The linear data structure (list) was used to store each of the incident tickets in sequential order. Each ticket was accessed using the list's index to perform operations such as addition, search, display, deletion, and count on the data structures.